

# HeatGeo: Heat-Diffusion Manifold Distillation for Compact Text Embedding Models

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## Abstract

Large text embedding models are central to semantic search, clustering, and retrieval-augmented generation (RAG), where quality depends on the geometry induced by the teacher embedding space. Existing embedding distillation methods provide useful supervision through teacher vectors, token or layer representations, pairwise preferences, and mini-batch relations. These signals, however, do not explicitly describe the intrinsic manifold structure encoded by the teacher’s neighborhoods: how local semantic regions connect, how similarity propagates through nearby examples, and which higher-order geometry should remain stable under compression. We propose **HeatGeo**, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo caches teacher embeddings, constructs a mutual  $k$ -nearest-neighbor graph as a discrete semantic manifold, and derives multi-scale diffusion distributions that describe how semantic mass flows across this manifold. The student matches these diffusion distributions over sampled candidate neighborhoods, with a single pointwise anchor to cached teacher embeddings as the only auxiliary term. We show that a separate spectral objective is unnecessary: because the heat kernel is a spectral low-pass filter, matching diffusion at several scales already transfers the low-frequency topology that eigenvector regression would supply, and the resulting two-term objective needs no global eigendecomposition. HeatGeo therefore turns the teacher’s final embeddings into geometry-aware supervision without requiring teacher hidden states, online teacher forward passes, or changes to the student architecture. We evaluate on STS, text classification, and pair classification to study whether manifold diffusion targets improve compact embedding distillation over pointwise, layer-level, and mini-batch relational baselines.

## Introduction

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. Their usefulness comes not only from individual vectors, but from the geometric structure those vectors induce: neighborhoods, semantic regions, and smooth paths between related examples. Strong recent embedding models, including LLM-based encoders, often learn rich semantic geometry but are costly to run at scale. Encoding large private collections, refreshing indexes as documents change, or serving retrieval on constrained hard-

ware therefore motivates compact student encoders that inherit the geometry of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, teacher-anchored layer alignment, and relational losses that match pairwise similarities within a mini-batch. These objectives provide valuable supervision at different granularities: individual examples, internal layers, sampled pairs, or local batch structure.

The geometric view suggests a complementary form of supervision. A teacher embedding space can be treated as a sampled semantic manifold: nearby examples define local tangent neighborhoods, multi-hop paths reveal semantic continuity, and low-frequency graph structure captures broader organization. Pointwise losses encourage coordinate agreement, layer-alignment methods improve internal transfer under capacity mismatch, and mini-batch relational losses preserve sampled similarity patterns. HeatGeo asks whether the student can also learn the teacher’s intrinsic manifold geometry through diffusion targets derived from the teacher embedding space.

This motivates the central question of this work: can a compact student inherit the teacher’s semantic manifold by matching heat-style diffusion on a teacher-induced neighborhood graph? The intuition is simple. If the teacher embeddings define a discrete manifold over training examples, then random walks on this manifold reveal how semantic mass flows from each anchor to nearby examples, broader semantic regions, and higher-order communities. Matching this diffusion process gives the student geometry-aware supervision that is not tied to teacher internals or a particular student architecture.

We propose **HeatGeo**, a heat-diffusion manifold distillation method for compact text embedding models. HeatGeo first caches teacher embeddings and constructs a mutual  $k$ -nearest-neighbor graph using teacher cosine similarity. This graph acts as a discrete approximation to the teacher’s se-

semantic manifold. From it, HeatGeo defines a transition matrix and precomputes multi-scale diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

HeatGeo adds exactly one auxiliary signal: a lightweight pointwise anchor that aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. We argue and verify empirically that a separate spectral objective, a natural candidate for transferring global topology, is redundant here. The heat kernel is a spectral low-pass filter, so matching diffusion at increasing scales already matches the graph spectrum with increasing smoothness; a truncated eigenvector-regression term distills the same structure a second time, while adding a global eigendecomposition to the precomputation.

Two design choices turn out to determine whether this objective transfers geometry at all. The graph temperature must be small enough that the diffusion targets stay far from uniform on their support, otherwise the loss degenerates into a binary neighbor/non-neighbor label. And the random walk must be lazy: a plain walk on a mutual  $k$ NN graph mixes within about two steps, so nominally distinct scales collapse onto the same distribution and the multi-scale objective silently becomes a single duplicated term.

Our contributions are:

- We formulate text embedding distillation as manifold geometry transfer, using the teacher embedding space as a discrete semantic manifold.
- We introduce HeatGeo, which distills multi-scale heat-diffusion distributions from a teacher-induced mutual  $k$ NN graph into a compact student encoder.
- We show that an explicit spectral anchoring term is redundant given multi-scale diffusion matching, yielding a two-term objective that requires no eigendecomposition, and we support this with an ablation that adds the term back.
- We identify graph temperature and walk laziness as the conditions under which multi-scale diffusion targets carry ranking information at all, and report the target-sharpness diagnostics that make this checkable before training.
- We provide an evaluation protocol for HeatGeo across semantic textual similarity, text classification, and pair classification benchmarks, with ablations for diffusion scale, auxiliary terms, candidate sampling, and curriculum.

## Related Work

**Text Embedding Models and Evaluation.** Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made

transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers and Gurevych 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao, Yao, and Chen 2021). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al. 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al. 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al. 2025), and Dewey targets long-context embedding for RAG-style retrieval (Zhang, Zou, and Zhou 2025). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should inherit the intrinsic geometry of the teacher embedding space. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills diffusion geometry induced by their neighborhoods.

## Distillation for Language and Embedding Models.

Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al. 2024; Ko et al. 2025; Anshumann et al. 2025; Kim et al. 2025; Xu et al. 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al. 2024b; Chen et al. 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al. 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al. 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic and Ruecksties 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al. 2026). These methods establish distilla-

tion as a central recipe for embedding model development. HeatGeo is complementary to this line: rather than introducing new teacher internals or task labels, it converts cached final embeddings into manifold diffusion targets.

**Local and Relational Supervision.** Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher’s absolute coordinates even when the student’s capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor across the teacher manifold. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al. 2019; Alqahtani et al. 2021; Dao et al. 2026). In particular, teacher-anchored layer alignment targets internal transfer and capacity mismatch, while HeatGeo targets the geometry induced by final teacher embeddings. These objectives operate on different aspects of distillation and can be viewed as complementary.

**Graph, Manifold, and Diffusion Perspectives.** Graph-based manifold learning studies how neighborhood graphs reveal lower-dimensional structure in high-dimensional data. Laplacian Eigenmaps use graph Laplacian eigenvectors to preserve local neighborhoods in low-dimensional representations (Belkin and Niyogi 2003), and Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman and Lafon 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al. 2025; Zhang et al. 2022). These works show that graph structure and diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph, and unlike Laplacian Eigenmaps we do not use spectral coordinates as an output at all: the diffusion-map view makes the eigenvectors implicit, since matching the heat kernel across scales is matching the spectrum under a low-pass filter. Instead, the graph is induced by a strong text embedding teacher and diffusion distributions become

soft manifold-neighborhood targets for the student. This gives compact text encoders an explicit geometry-aware supervision signal derived from the teacher embedding manifold, without a global eigendecomposition.

## Problem Setup

Let  $\mathcal{D} = \{x_i\}_{i=1}^N$  denote a training corpus. A teacher encoder  $T$  maps each input to a teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \quad (1)$$

and a compact student encoder  $S_\theta$  maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \quad (2)$$

The goal is to train  $S_\theta$  so that the student embedding space preserves the semantic geometry induced by the teacher, especially nearest-neighbor relations, local semantic regions, and higher-order manifold structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N} \sum_{i=1}^N (1 - \cos(W s_i, t_i)), \quad (3)$$

where  $W$  is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch  $B$ :

$$\mathcal{L}_{\text{rel}} = \|G_B^S - G_B^T\|_F^2, \quad (4)$$

where  $G_B^S$  and  $G_B^T$  are cosine-similarity matrices over examples in  $B$ . HeatGeo lifts the relational view from mini-batch snapshots to a teacher-induced semantic manifold represented by a neighborhood graph.

## HeatGeo

### Teacher Semantic Graph

HeatGeo first encodes the full training corpus with the teacher and constructs a teacher semantic graph. For each example  $x_i$ , let

$$\mathcal{N}_k^T(i) = \text{TopK}_{j \neq i} \cos(t_i, t_j) \quad (5)$$

be its top- $k$  teacher nearest neighbors. To reduce noisy one-directional edges and hubness, HeatGeo uses a mutual  $k$ NN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \quad (6)$$

The weighted adjacency matrix  $W^T$  is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where  $\tau_g$  controls graph sharpness. Let  $D^T$  be the diagonal degree matrix. The row-normalized transition matrix is

$$P^T = (D^T)^{-1} W^T. \quad (8)$$

Each row  $P_i^T$  defines a one-step semantic random-walk distribution from anchor  $x_i$ .

The graph temperature  $\tau_g$  is not a cosmetic hyperparameter. A mutual  $k$ NN graph has degree of order  $k$ , and teacher cosine similarities among the top- $k$  neighbors of an anchor occupy a narrow band. If  $\tau_g$  is large relative to that band, every row of  $P^T$  is close to uniform over its support, and the diffusion targets below degrade into binary neighbor/non-neighbor labels that carry no ranking information. We therefore report  $D_{\text{KL}}(p_{i,r}^T \| \mathcal{U}_{\text{supp}})$ , the divergence of each target from the uniform distribution on its own support, as a build-time diagnostic and select  $\tau_g$  so that this quantity stays well above zero.

### Multi-Scale Heat Diffusion Targets

A one-step transition captures direct neighbors, but embedding spaces also contain higher-order semantic structure. HeatGeo therefore uses multiple diffusion scales of a *lazy* random walk,

$$Q^T = \frac{1}{2} (I + P^T), \quad P_r^T = (Q^T)^r, \quad r \in \mathcal{R}, \quad (9)$$

where  $\mathcal{R}$  is typically  $\{1, 2, 4\}$ . Small  $r$  preserves sharp local neighborhoods, while larger  $r$  captures semantic regions and multi-hop communities. Laziness is what makes  $\mathcal{R}$  a graded family: a plain walk on a mutual  $k$ NN graph mixes within its local component in about two steps, so  $(P^T)^2$  and  $(P^T)^4$  collapse onto nearly the same distribution and the additional scales contribute duplicated terms rather than new structure. The lazy walk retains mass at intermediate step counts and yields scales that remain distinguishable. In implementation, diffusion is approximated sparsely by propagating probability mass through graph neighbor lists and retaining top candidates, with a single walk snapshotted at each scale in  $\mathcal{R}$ .

For each anchor  $x_i$ , HeatGeo builds a candidate set

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \quad (10)$$

where  $C_i^+$  contains high-probability diffusion neighbors,  $C_i^h$  contains teacher hard negatives, and  $C_i^r$  contains random negatives, each filling a fixed slot quota. Diffusion positives are drawn round-robin across the scales in  $\mathcal{R}$ , so that every scale contributes candidates on which it can be scored; filling the quota scale by scale instead lets the sharpest scale exhaust the budget and leaves the broader scales without support. The teacher diffusion target at scale  $r$  is the diffusion mass restricted and renormalized over  $C_i$ :

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \quad j \in C_i. \quad (11)$$

This target is a soft distribution over semantic relatedness, rather than a binary positive-negative label.

### Student Neighborhood Distribution

Given an anchor embedding  $s_i$  and candidate embeddings  $\{s_j : j \in C_i\}$ , the student induces a distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_s)}, \quad j \in C_i, \quad (12)$$

where  $\tau_s$  is a student temperature. The main diffusion distillation loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(p_{i,r}^T \| p_i^S), \quad (13)$$

where  $B$  is a mini-batch and  $\omega_r$  is the scale weight.

### Why No Separate Spectral Term

A natural addition would be a spectral term that regresses student embeddings onto the leading eigenvectors of the graph Laplacian  $L^T = I - (D^T)^{-1/2} W^T (D^T)^{-1/2}$ , on the grounds that diffusion matching is local while eigenvectors describe global topology. We deliberately omit such a term, because it distills information that  $\mathcal{L}_{\text{diff}}$  already carries. The heat kernel admits the spectral expansion

$$e^{-tL^T} = \sum_{\ell} e^{-t\lambda_{\ell}} u_{\ell} u_{\ell}^T, \quad (14)$$

so matching diffusion at several scales is matching the graph spectrum under a low-pass filter whose bandwidth is set by  $t$ . A truncated eigenvector-regression term is a strictly weaker version of the large-scale member of  $\mathcal{R}$ , and adding it distills the same structure twice. Omitting it also removes the only global eigendecomposition from the pipeline, leaving a method whose precomputation is a  $k$ NN search followed by sparse propagation. Section reports the ablation that adds the term back.

### Training Objective

HeatGeo includes a lightweight pointwise anchor for stability:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|} \sum_{i \in B} (1 - \cos(W_a s_i, t_i)), \quad (15)$$

where  $W_a$  maps student embeddings to the teacher dimension. This term supplies the pointwise component that a purely relational objective leaves unconstrained, and it is the only auxiliary HeatGeo uses.

The final objective is therefore two terms with a single balancing weight:

$$\mathcal{L}_{\text{HeatGeo}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}}. \quad (16)$$

All scales in  $\mathcal{R}$  are active from the first step. We also evaluated a curriculum that introduces scales one epoch at a time; because the scale weights are renormalized over the active set, such a curriculum monotonically shifts weight away from the sharpest local scale as training proceeds, and we found no benefit from it (Section ).

Since the loss value alone cannot distinguish a student that has learned the teacher’s neighborhood ranking from one whose embeddings have collapsed toward a uniform candidate distribution, we log the entropy of  $p_i^S$  relative to  $\log |C_i|$  and the student probability mass falling on the teacher’s top-ranked candidates.

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**Algorithm 1** HeatGeo Training

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**Require:** Corpus  $\mathcal{D}$ , teacher  $T$ , student  $S_\theta$ , graph size  $k$ , diffusion scales  $\mathcal{R}$

- 1: Cache teacher embeddings  $t_i = T(x_i)$  for all  $x_i \in \mathcal{D}$
- 2: Build a mutual  $k$ NN teacher graph  $W^T$
- 3: Compute transition matrix  $P^T = (D^T)^{-1}W^T$  and lazy walk  $Q^T = \frac{1}{2}(I + P^T)$
- 4: Precompute sparse diffusion candidates and targets  $p_{i,r}^T$  from one walk snapshot at each  $r \in \mathcal{R}$
- 5: **for** each training epoch **do**
- 6:   **for** each mini-batch  $B$  **do**
- 7:     Encode anchors and candidate texts with  $S_\theta$
- 8:     Compute student candidate distributions  $p_i^S$
- 9:     Compute  $\mathcal{L}_{\text{diff}}$  and  $\mathcal{L}_{\text{anchor}}$
- 10:    Update  $\theta$  by minimizing  $\mathcal{L}_{\text{HeatGeo}}$
- 11:   **end for**
- 12: **end for**
- 13: **return** Distilled student encoder  $S_\theta$

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## Experiments

### Experimental Setup

**Models.** Our implementation uses a compact BERT/MiniLM-style student encoder and an LLM-based teacher embedding model. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher model is removed from GPU memory; HeatGeo training uses only cached teacher embeddings and precomputed diffusion targets.

**Training Data.** We train on text-pair data assembled from multiple semantic tasks. Each training row provides either a text pair or a single text field depending on the task format. HeatGeo constructs the teacher graph over the training examples, using the anchor text for candidate neighborhoods.

**Evaluation Tasks and Metrics.** We evaluate the distilled student on three task families. For semantic textual similarity, we report Spearman correlation on SICK, STS12, and STS-B. For text classification, we freeze the encoder, train a logistic regression classifier on top of embeddings, and report accuracy and macro-F1 on Banking77, Emotion, and Tweet. For pair classification, we threshold cosine similarity and report accuracy, macro-F1, precision, recall, and average precision on MRPC, SciTail, and WiC.

**Baselines.** We compare HeatGeo with the undistilled student, pointwise teacher anchoring, mini-batch relational distillation, cross-tokenizer alignment methods such as CDM and DSKD, EMO-style token/attention alignment, Stella-style multi-faceted embedding distillation, TALAS teacher-anchored layer alignment, and the full teacher embedding model when feasible. These baselines cover output-level, layer-level, tokenizer-aware, and relational distillation signals.

**Implementation Details.** Unless otherwise specified, HeatGeo uses graph size  $k = 50$ , graph temperature  $\tau_g = 0.05$ , lazy diffusion scales  $\{1, 2, 4\}$ , candidate size

64, 32 diffusion positives, 16 hard negatives, 16 random negatives, and five training epochs. The scale weights are (1.0, 0.5, 0.25), giving higher weight to sharper local diffusion, and the student temperature is  $\tau_s = 0.07$ . The objective has a single balancing weight,  $\lambda_{\text{anchor}} = 0.05$ .

**Target Sharpness Diagnostics.** Because the informativeness of the diffusion targets depends on the teacher and the corpus, we report  $D_{\text{KL}}(p_{i,r}^T \| \mathcal{U}_{\text{supp}})$  per scale and the pairwise divergence between consecutive scales at graph-construction time. These two numbers reveal, before any training, whether the target carries ranking information and whether the scales are genuinely distinct;  $\tau_g$  was selected on this basis rather than by downstream search.

### Main Results

Table 1 reports downstream embedding quality across the evaluation tasks. The key comparison is whether manifold diffusion supervision improves compact embedding distillation relative to output-level, layer-level, and mini-batch relational baselines while retaining the efficiency of the compact student.

### Ablation Study

We isolate the contributions of HeatGeo with the ablations in Table 2. Removing diffusion reduces the method to pointwise anchoring alone, and removing the anchor leaves a purely relational objective. Two ablations justify what the objective leaves out: adding a spectral term back tests the redundancy argument of Section , and adding an InfoNCE task term tests whether contrastive supervision contributes anything beyond the diffusion targets. The remaining rows probe the design choices that make the multi-scale objective non-degenerate: the plain walk against the lazy walk, larger  $\tau_g$  against the selected value, the scale curriculum against activating all scales from the start, and the role of hard and random negatives.

### Discussion

HeatGeo is designed around the observation that teacher embeddings define more than isolated targets: they define a semantic manifold whose diffusion geometry can serve as a distillation signal. The method has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that HeatGeo requires a graph pre-processing step before student training. Building a teacher graph is worthwhile when the same training data supports substantial distillation, but it adds overhead compared with purely mini-batch objectives. HeatGeo also depends on the quality and coverage of the teacher graph: if the teacher embeddings contain spurious neighborhoods, diffusion may propagate those errors. Future work can study approximate

Table 1: Experimental results on nine datasets. Datasets marked with  $\star$  are in-domain, while the remaining datasets are out-of-domain.

|                                 | Classification (F1) |       |          | Pair Classification (AP) |         |       | STS (Spearman) |       |       | Domain Avg |         | Avg.  |
|---------------------------------|---------------------|-------|----------|--------------------------|---------|-------|----------------|-------|-------|------------|---------|-------|
|                                 | Banking77           | Tweet | Emotion* | MRPC                     | SciTail | WiC*  | SICK           | STS12 | STSB* | Avg-In     | Avg-Out |       |
| BGE-M3 → BERT-base              |                     |       |          |                          |         |       |                |       |       |            |         |       |
| Teacher                         | 93.52               | 73.85 | 68.56    | 85.81                    | 91.87   | 61.47 | 79.18          | 78.73 | 84.87 | 71.63      | 83.83   | 79.76 |
| Student base                    | 84.86               | 47.79 | 68.02    | 77.36                    | 67.74   | 59.22 | 42.43          | 21.54 | 20.30 | 42.44      | 60.33   | 54.36 |
| SimCSE-unsup                    | 89.21               | 69.13 | 53.55    | 82.86                    | 76.59   | 71.64 | 68.19          | 61.68 | 70.36 | 65.18      | 74.61   | 71.47 |
| CDM                             | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| DSKD                            | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| Jasper and Stella               | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| DistillCSE                      | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| EMO                             | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| TALAS                           | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| HeatGeo                         | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| Qwen-Embedding-0.6B → BERT-base |                     |       |          |                          |         |       |                |       |       |            |         |       |
| Teacher                         | 93.21               | 71.33 | 66.53    | 83.37                    | 90.00   | 66.85 | 80.64          | 77.11 | 84.60 | 72.66      | 82.61   | 79.29 |
| Student base                    | 84.86               | 47.79 | 68.02    | 77.36                    | 67.74   | 59.22 | 42.43          | 21.54 | 20.30 | 42.44      | 60.33   | 54.36 |
| SimCSE-unsup                    | 89.21               | 69.13 | 53.55    | 82.86                    | 76.59   | 71.64 | 68.19          | 61.68 | 70.36 | 65.18      | 74.61   | 71.47 |
| CDM                             | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| DSKD                            | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| Jasper and Stella               | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| DistillCSE                      | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| EMO                             | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| TALAS                           | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| HeatGeo                         | —                   | —     | —        | —                        | —       | —     | —              | —     | —     | —          | —       | —     |
| Qwen-Embedding-4B → BERT-base   |                     |       |          |                          |         |       |                |       |       |            |         |       |
| Teacher                         | 93.63               | 72.22 | 68.94    | 84.79                    | 91.77   | 66.80 | 83.43          | 82.55 | 86.87 | 74.20      | 84.73   | 81.22 |
| Student base                    | 84.86               | 47.79 | 68.02    | 77.36                    | 67.74   | 59.22 | 42.43          | 21.54 | 20.30 | 42.44      | 60.33   | 54.36 |
| SimCSE-unsup                    | 89.21               | 69.13 | 53.55    | 82.86                    | 76.59   | 71.64 | 68.19          | 61.68 | 70.36 | 65.18      | 74.61   | 71.47 |
| CDM                             | 89.75               | 70.58 | 53.11    | 84.83                    | 75.19   | 71.06 | 69.37          | 68.48 | 73.92 | 66.03      | 76.37   | 72.92 |
| DSKD                            | 89.66               | 69.93 | 54.51    | 83.13                    | 77.95   | 71.53 | 68.65          | 63.37 | 71.42 | 65.82      | 75.45   | 72.24 |
| Jasper and Stella               | 89.71               | 73.83 | 65.98    | 85.33                    | 80.99   | 70.50 | 73.96          | 67.90 | 75.65 | 70.71      | 78.62   | 75.98 |
| DistillCSE                      | 90.94               | 70.90 | 60.80    | 82.86                    | 78.98   | 68.63 | 75.12          | 72.61 | 77.08 | 68.84      | 78.57   | 75.32 |
| EMO                             | 90.78               | 73.54 | 67.48    | 84.46                    | 81.57   | 69.76 | 75.66          | 68.83 | 77.17 | 71.47      | 79.14   | 76.58 |
| TALAS                           | 92.34               | 73.74 | 64.41    | 85.97                    | 82.66   | 68.65 | 78.38          | 72.27 | 79.98 | 71.01      | 80.89   | 77.60 |
| HeatGeo                         | 92.62               | 72.27 | 66.42    | 83.57                    | 89.15   | 66.43 | 79.69          | 70.43 | 81.79 | 71.55      | 81.29   | 78.04 |

nearest-neighbor construction, adaptive diffusion scales, and online graph refresh for changing data.

## Conclusion

We introduced HeatGeo, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo treats cached teacher embeddings as samples from a semantic manifold and distills multi-scale lazy diffusion over a teacher-induced mutual  $k$ NN graph, with a single pointwise anchor as its only auxiliary term. Because the heat kernel is itself a spectral low-pass filter, this two-term objective transfers low-frequency topology without a separate eigenvector-regression loss and without any global eigendecomposition. This gives the student a geometry-aware supervision signal while requiring only cached teacher embeddings and no student architecture changes. HeatGeo is therefore a practical manifold-oriented complement to output-level, layer-level, and mini-batch relational embedding distillation objectives.

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Table 2: Ablation template for HeatGeo. Report average score across task groups or a selected validation suite. The first block ablates the two terms of the objective, the second tests whether the terms we deliberately excluded would add anything, and the third probes the conditions under which the multi-scale targets are non-degenerate.

| Variant                                            | Average Score |
|----------------------------------------------------|---------------|
| HeatGeo full                                       | —             |
| w/o diffusion loss $\mathcal{L}_{\text{diff}}$     | —             |
| w/o pointwise anchor $\mathcal{L}_{\text{anchor}}$ | —             |
| + spectral anchoring term                          | —             |
| + InfoNCE task term                                | —             |
| plain walk instead of lazy walk                    | —             |
| $\tau_g = 0.1$ instead of 0.05                     | —             |
| + diffusion-scale curriculum                       | —             |
| w/o hard negatives                                 | —             |
| w/o random negatives                               | —             |
| single scale $r = 1$                               | —             |
| single scale $r = 4$                               | —             |

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